

Derivatives and Rates of Change

Section 2.1

A. Slope of Secant Functions

Recall: Slope = $m = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$. From this we are able to derive:

Slope of the Secant Line to a Function: $m = \frac{y_2 - y_1}{x_2 - x_1}$ or $m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

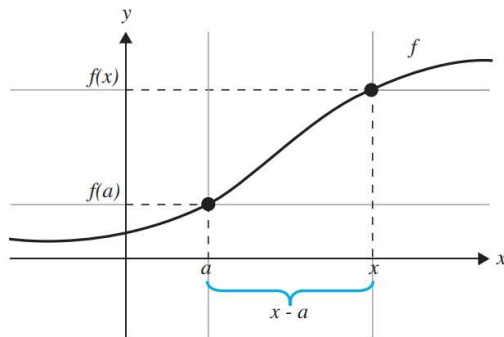
- (video) a.** Find the **slope** of the secant line to the function $f(x) = x^2 + 3x - 5$ between $x = 1$ and $x = 2$.

b. (video) Find the **equation** of the secant line to the function $f(x) = x^2 + 3x - 5$ between $x = 1$ and $x = 2$.
- a.** Find the **slope** of the secant line to the function $f(x) = \sqrt{x}$ between $x = 1$ and $x = 9$.

- b. Find the **equation** of the secant line to the function $f(x) = \sqrt{x}$ between $x = 1$ and $x = 9$.
3. Estimate the slope of the **tangent** line to the function $y = x^2$ at the point $(1, 1)$ by calculating the slope of the **secant** line between $x = 1$ and $x = 1.1$, between $x = 1$ and $x = 1.01$, and between $x = 1$ and $x = 1.001$.

A generalization of the previous example gives the definition: $\text{Slope} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$. Similarly we can approach this by defining the distance between x and a as $x - a = h$.

Slope of the tangent line of $f(x)$ at point specific a



In order to get the two points P_1 and P_2 as close together as possible, we need the space $x - a \rightarrow 0$.

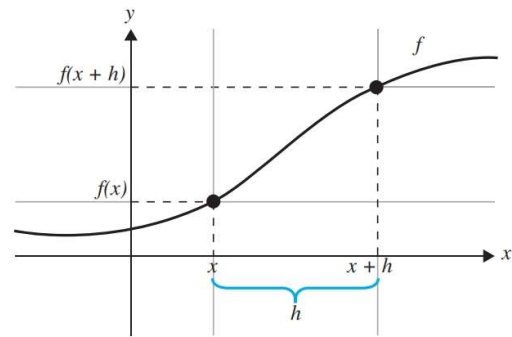
So, the slope between P_1 and P_2 is:

$$m = \frac{\Delta y}{\Delta x} = \frac{f(x) - f(a)}{x - a}$$

but as the space $x - a \rightarrow 0$, we have

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Slope of the tangent line of $f(x)$ at a general point x



In order to get the two points P_1 and P_2 as close together as possible, we need the space $h \rightarrow 0$.

So, the slope between P_1 and P_2 is:

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{(x+h) - (x)} \\ &= \frac{f(x+h) - f(x)}{h} \end{aligned}$$

but as the space $h \rightarrow 0$, we have

$$m = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

B. Definition of Derivative

The definition of a derivative (aka the slope of the tangent function) is given as:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{or} \quad f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

*Note: In this section, we will use the DEFINITION OF THE DERIVATE to calculate all derivatives. (This means we will be doing it the *long way*!)

Examples

4. **(video)** Find the equation of the tangent line to the function $f(x) = x^2 + 3x + 10$ where $x = 5$.

5. **(video)** Find the derivative of the function $f(x) = x^2 + 3x + 10$ using the difference quotient and the definition of derivative.

6. Find the equation of the tangent line to the function $f(x) = x^2 - 3x + 1$ where $x = 5$.
7. A person standing on top of a 200 ft. tall building throws a ball into the air with a velocity of 96 ft/sec. The function $s(t) = -16t^2 + 96t + 200$ gives the ball's height above ground, t seconds after it was thrown. Find the instantaneous velocity of the ball at $t = 2$ seconds

8. The position of a particle is given by the values of the tables

$t(\text{seconds})$	0	1	2	3	4	5
$s(\text{feet})$	0	14	47	51	86	103

Find the average velocity for the time period beginning when $t = 2$ and lasting

1. 3 s (i.e., for the time interval $[2,5]$)
 2. 2 s
 3. 1 s
9. a. The equation of the tangent line to the graph of $y = g(x)$ at $x = 3$ if $g(3) = -3$ and $g'(3) = 7$ is $y =$
- b. If the tangent line to $y = f(x)$ at $(2,10)$ passes through the point $(0,4)$,
- then $f(2) =$
- and $f'(2) =$

10. $\lim_{h \rightarrow 0} \frac{\sqrt{81+h} - 9}{h}$ represents the derivative of the function $f(x) = \underline{\hspace{2cm}}$ at the number $a = \underline{\hspace{2cm}}$

11. $\lim_{x \rightarrow 6} \frac{2^x - 64}{x - 6}$ represents the derivative of the function $f(x) = \underline{\hspace{2cm}}$ at the number $a = \underline{\hspace{2cm}}$